

Urban Sprawl Monitoring and Future Prediction in Southern Indian Metropolises using GHSL, Sentinel-2 and Machine Learning



VIT
CHENNAI



SET Conference Final Review
MCA - PSET695J

Presented by

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1. ABSTRACT

- Rapid urbanization in Indian metropolitan cities poses serious challenges for sustainable development, infrastructure planning, and environmental management. This project presents a scalable, cloud-based geospatial framework for monitoring and predicting urban sprawl across three major southern Indian cities: Chennai, Hyderabad, and Bengaluru from 2000 to 2025, with future predictions for 2030.
- The study leverages Global Human Settlement Layer (GHSL) data for 2000, 2010, 2020, and 2025, processed on Google Earth Engine (GEE). Sentinel-2 Level-2A imagery was used for high-resolution built-up classification using spectral indices (NDBI, NDVI, MNDWI) and a supervised Random Forest algorithm. Two machine learning models: Polynomial Regression and Random Forest were applied pixel-wise to predict the 2030 urban extent.
- Results reveal significant built-up expansion in all three cities, with Chennai and Hyderabad showing predicted growth of 27–32 km² by 2030. Model validation achieved R² scores up to 0.83, confirming reliability. This framework supports evidence-based urban planning and environmental policy decisions.

2. INTRODUCTION

- Urbanization is accelerating rapidly across Indian metropolises, leading to unplanned, low-density, and fragmented urban expansion commonly known as urban sprawl. Southern India's tech-hub cities Chennai, Hyderabad, and Bengaluru are among the fastest-growing cities globally.
- Conventional monitoring methods (field surveys, census analysis) are slow and spatially limited. This project addresses this gap using Remote Sensing and GIS specifically GHSL satellite data and Sentinel-2 imagery processed via Google Earth Engine (GEE). A new add-on in this review cycle is the integration of Polynomial Regression and Random Forest ML models to predict 2030 urban extent, providing actionable insights for city planners and policymakers.

3. LITERATURE SURVEY

S.No	Article Title	Techniques Used	Key Findings	Limitations
1	Baqa et al. (2021) "Monitoring and Modelling of Urban Sprawl Using Remote Sensing and GEE"	Landsat + Random Forest on Google Earth Engine	RF classifier achieved high accuracy in mapping Karachi's sprawl; identified major expansion corridors from 2000–2020	Single-city focus; no future prediction beyond observed period
2	Wang et al. (2020) "Urban Expansion Monitoring Using CART Classification on GEE" (Huizhou, China)	CART decision tree on GEE platform, Landsat time series	Detected 3× increase in built-up area over two decades; GEE enabled scalable processing	CART less accurate than ensemble models; limited to one city
3	Seto et al. (2011) "Global Forecasts of Urban Expansion to 2030 and Direct Impacts on Biodiversity"	Remote sensing + logistic regression, global LULC data	Urban land area projected to triple by 2030 globally; biodiversity hotspots at severe risk	Resolution too coarse (1 km ²) for city-level planning
4	Boori et al. (2021) "Urban Growth Analysis Using Google Earth Engine"	Spatial metrics, multi-temporal Landsat, GEE	Identified radial and ribbon growth patterns in multiple Indian cities; GEE proved efficient for bulk processing	No spectral index computation; limited to visual analysis

LITERATURE SURVEY (CONTD..)

S.No	Article Title	Techniques Used	Key Findings	Limitations
5	Tariq et al. (2024) "Spatio-temporal Analysis of Urban Expansion Using GEE and Predictive Models" (<i>Scientific Reports</i>)	Landsat + RF classification on GEE; CA-ANN for 2030 forecast	Built-up area in Multan and Sargodha doubled from 1990–2020; ANN-CA projected further 40% growth by 2030	CA-ANN requires extensive transition data; complex calibration
6	Mhangara et al. (2024) "Urban Sprawl Dynamics Using ML, CA-Markov and Shannon Entropy" (<i>Environmental Systems Research</i>)	CA-Markov chain, Shannon entropy, ML classification	Quantified sprawl intensity across periods; Shannon entropy confirmed dispersed growth pattern	Shannon entropy does not capture spatial distribution of new growth
7	Roy et al. (2024) "Analysis of Urban Sprawl Growth and Prediction Using Remote Sensing and ML" (<i>ScienceDirect</i>)	Landsat + Shannon entropy + ANN-MLP + CA, QGIS MOLUSCE plugin	Built-up area projected to grow from 51.64 km ² (2024) to 62.31 km ² by 2034 in Vila Velha, Brazil	Prediction limited to one city; ANN-MLP needs large labelled datasets
8	Sakti et al. (2024) "Machine Learning Based Urban Sprawl Assessment Using Integrated Multi-hazard Impact" (<i>Scientific Reports</i>)	Long-term RS data, ML, urban sprawl priority index	First study to combine economic, environmental, and disaster risk into a sprawl priority index	Computationally heavy; not transferable without large local datasets

LITERATURE SURVEY (CONTD..)

S.No	Article Title	Techniques Used	Key Findings	Limitations
9	Siddiqui et al. (2025) "Ensemble ML for Predicting Urban Expansion in Lucknow, India" (<i>Environmental Earth Sciences</i>)	XGBoost, CatBoost, LightGBM + Landsat + SHAP explainability	XGBoost achieved 92.4% accuracy (AUC 0.97); proximity to roads and CBD most influential features	Computationally intensive; requires rich socio-economic driver data
10	Pan et al. (2024) "Role of Random Forest and Markov Chain in Understanding Metropolitan Urban Growth" (<i>Frontiers in Forests</i>)	RF classification + CA-Markov on GEE, 1991–2051 forecast	RF outperformed ANN for LULC classification; Markov chain extended predictions to 2051	Markov chain assumes static transition probabilities; doesn't adapt to policy change

4. RESEARCH GAPS

- Existing studies focus on **single cities**; no comparative multi-city sprawl analysis across southern India
- Limited use of **2025 GHSL data** most studies stop at 2020
- Few studies combine **GHSL + Sentinel-2** for cross-validation of built-up detection
- **Future prediction (2030)** is largely absent in Indian urban sprawl literature
- Most ML approaches are classification-only; pixel-wise **spatiotemporal regression for forecasting** is underexplored
- Lack of open, reproducible, **cloud-based (GEE) pipelines** for urban monitoring in India

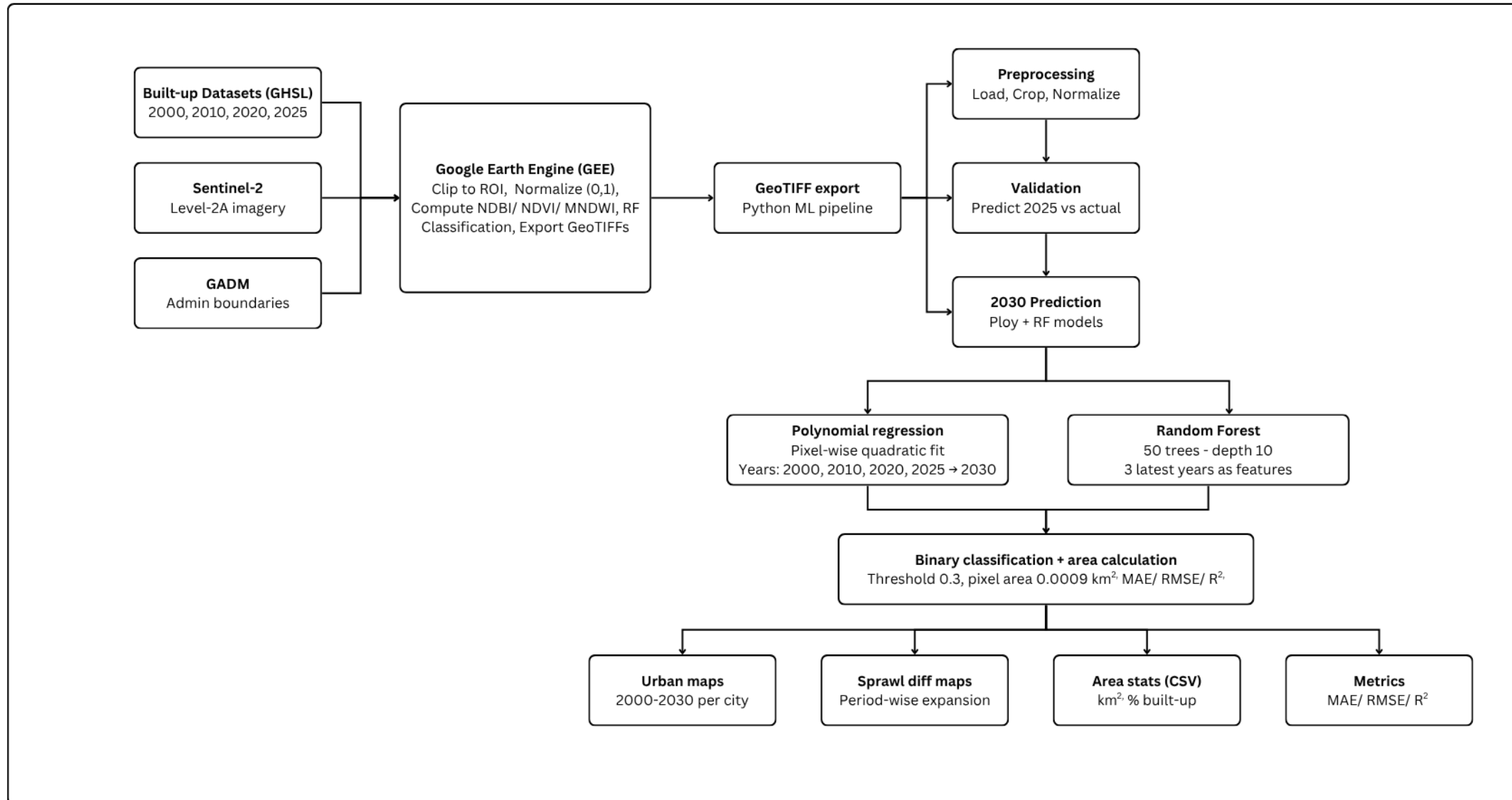
5. OBJECTIVES

- Monitor built-up area expansion in Chennai, Hyderabad, and Bengaluru from 2000 to 2025 using GHSL data
- Classify urban land cover for 2024 using Sentinel-2 imagery with NDBI, NDVI, MNDWI indices and Random Forest
- Quantify urban sprawl across three decadal periods (2000–2010, 2010–2020, 2020–2025)
- Develop pixel-wise Polynomial and Random Forest regression models to predict 2030 urban extent
- Compare model performance using MAE, RMSE, and R^2 metrics
- Provide a scalable, reproducible GEE-based framework for urban sprawl monitoring

6. MODULE IDENTIFICATION

- **Module 1: Data Acquisition** GHSL built-up layers (2000–2025) and Sentinel-2 Level-2A imagery loaded into Google Earth Engine; administrative boundaries from GADM
- **Module 2: Preprocessing** Cloud masking, atmospheric correction, normalization of GHSL to 0–1 scale, image cropping to consistent dimensions
- **Module 3: Spectral Index Computation** NDBI, NDVI, MNDWI calculated from Sentinel-2 bands for built-up detection and classification
- **Module 4: Urban Classification** Supervised Random Forest classifier trained on digitized samples (built-up, vegetation, water, bare land)
- **Module 5: Sprawl Quantification** Difference maps computed between years; built-up area and gain statistics calculated in km²
- **Module 6: Prediction (2025 & 2030)** Polynomial Regression and Random Forest Regression models trained pixel-wise on GHSL time series to predict future urban extent
- **Module 7: Visualization & Export** Multi-year urban maps, sprawl difference maps, model comparison outputs, and CSV statistics exported

7. SYSTEM ARCHITECTURE



8. DEVELOPMENT & INTEGRATION

- **Phase 1: GEE Implementation:** GHSL images for 2000, 2010, 2020 loaded from JRC collection; 2025 data uploaded as custom asset. Images normalized to 0-1 scale, clipped to city boundaries, and exported as 30m GeoTIFFs.
- **Phase 2: Python Processing:** GeoTIFFs loaded via Rasterio; images aligned by cropping to minimum common dimensions. Threshold of 0.3 used for binary urban classification. Polynomial regression fit using NumPy least squares; Random Forest (50 estimators) trained via Scikit-learn.
- **Phase 3: 2030 Prediction:** Polynomial model uses years [2000, 2010, 2020, 2025] as features with quadratic terms. RF model uses pixel values from the 3 most recent years as input features to predict the next timestep.
- **Challenges & Solutions:** Mismatched image dimensions resolved by cropping to minimum rows/cols; NaN values masked before training and metric computation.

9. PROTOTYPE / RESEARCH FRAMEWORK

Input Data: GHSL Built-up Layers (2000, 2010, 2020, 2025) at 30m resolution + Sentinel-2 2024 imagery

Framework Steps:

- Load & normalize multi-temporal GHSL stack on GEE
- Compute NDBI/NDVI/MNDWI from Sentinel-2; train RF classifier for 2024 built-up map
- Export city-specific GeoTIFFs for all years
- Train Polynomial + RF regression models in Python on historical pixel values
- Predict 2025 (validation) and 2030 (forecast) urban maps
- Compute binary urban masks (threshold 0.3), calculate expansion areas in km²
- Visualize multi-year comparison maps and model outputs

Cities: Bengaluru · Hyderabad · Chennai

10. OUTCOME

Predicted 2030 Urban Expansion Areas

City	Polynomial Model (km²)	Random Forest (km²)
Bengaluru	18.53	28.83
Hyderabad	25.46	18.46
Chennai	27.46	31.64

- Multi-year urban growth maps generated for all three cities (2000, 2010, 2020, 2025, 2030)
- Sprawl difference maps showing new urban areas clearly identified
- GEE pipeline successfully processes and exports all city data at 30m resolution

11. RESULTS

Model Validation

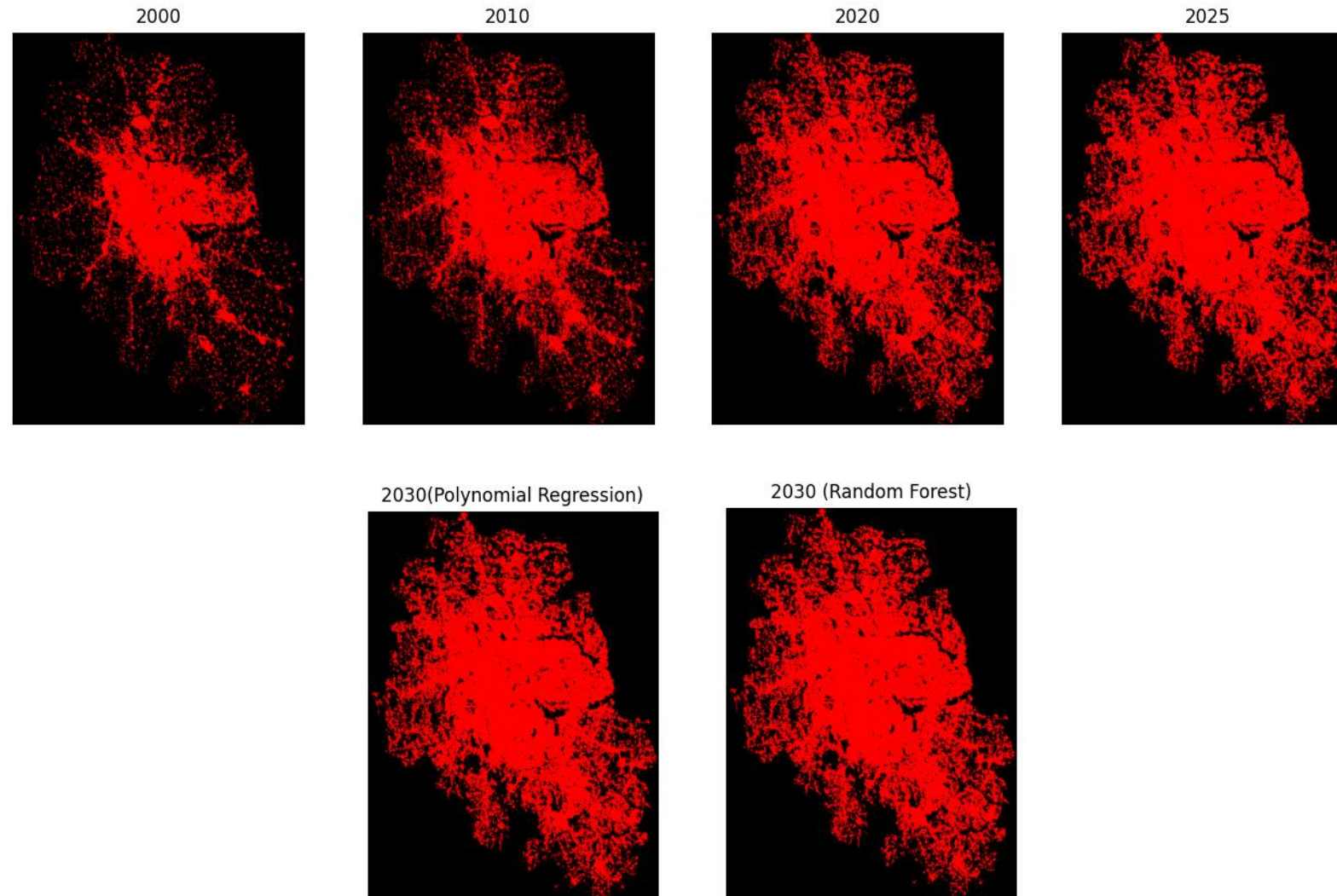
City	Model	MAE	RMSE	R ²
Bengaluru	Polynomial	0.120	0.285	0.631
Bengaluru	Random Forest	0.119	0.237	0.744
Hyderabad	Polynomial	0.064	0.214	0.793
Hyderabad	Random Forest	0.077	0.191	0.834
Chennai	Polynomial	0.053	0.199	0.792
Chennai	Random Forest	0.072	0.185	0.822

Key Observations:

- Random Forest outperforms the Polynomial model in R² across all cities
- Hyderabad RF model achieves the highest R² (0.834)
- All models validated against actual 2025 GHSL data
- Binary classification threshold of 0.3 optimized for built-up detection

11. RESULTS

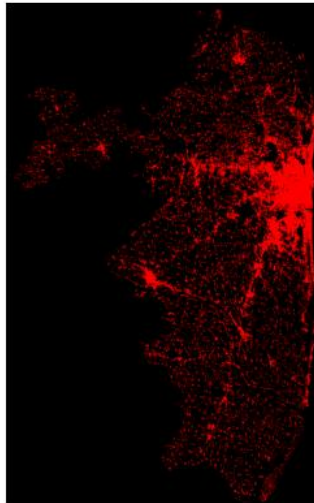
Urban Sprawl in Bengaluru



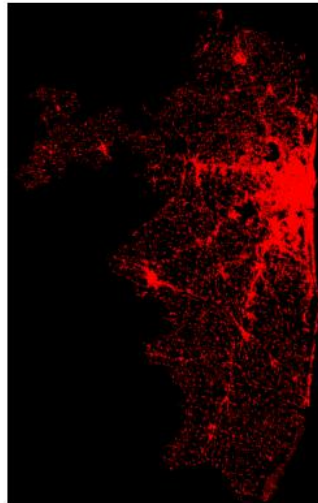
11. RESULTS

Urban Sprawl in Chennai

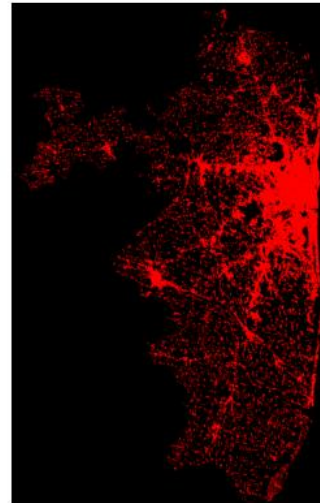
2000



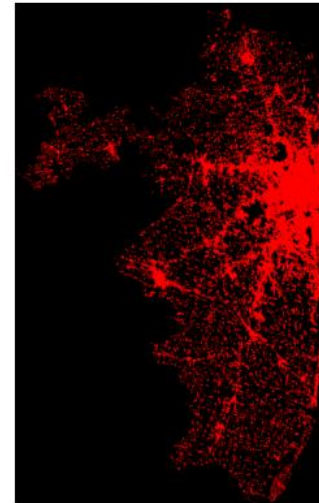
2010



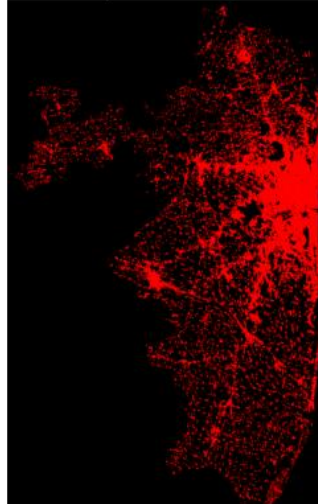
2020



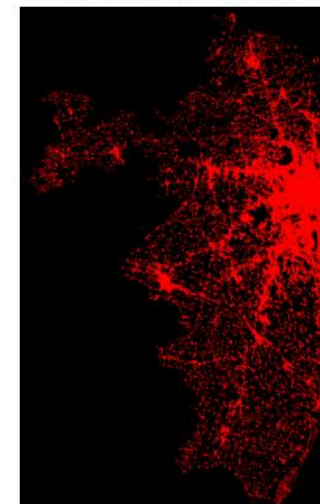
2025



2030(Polynomial Regression)

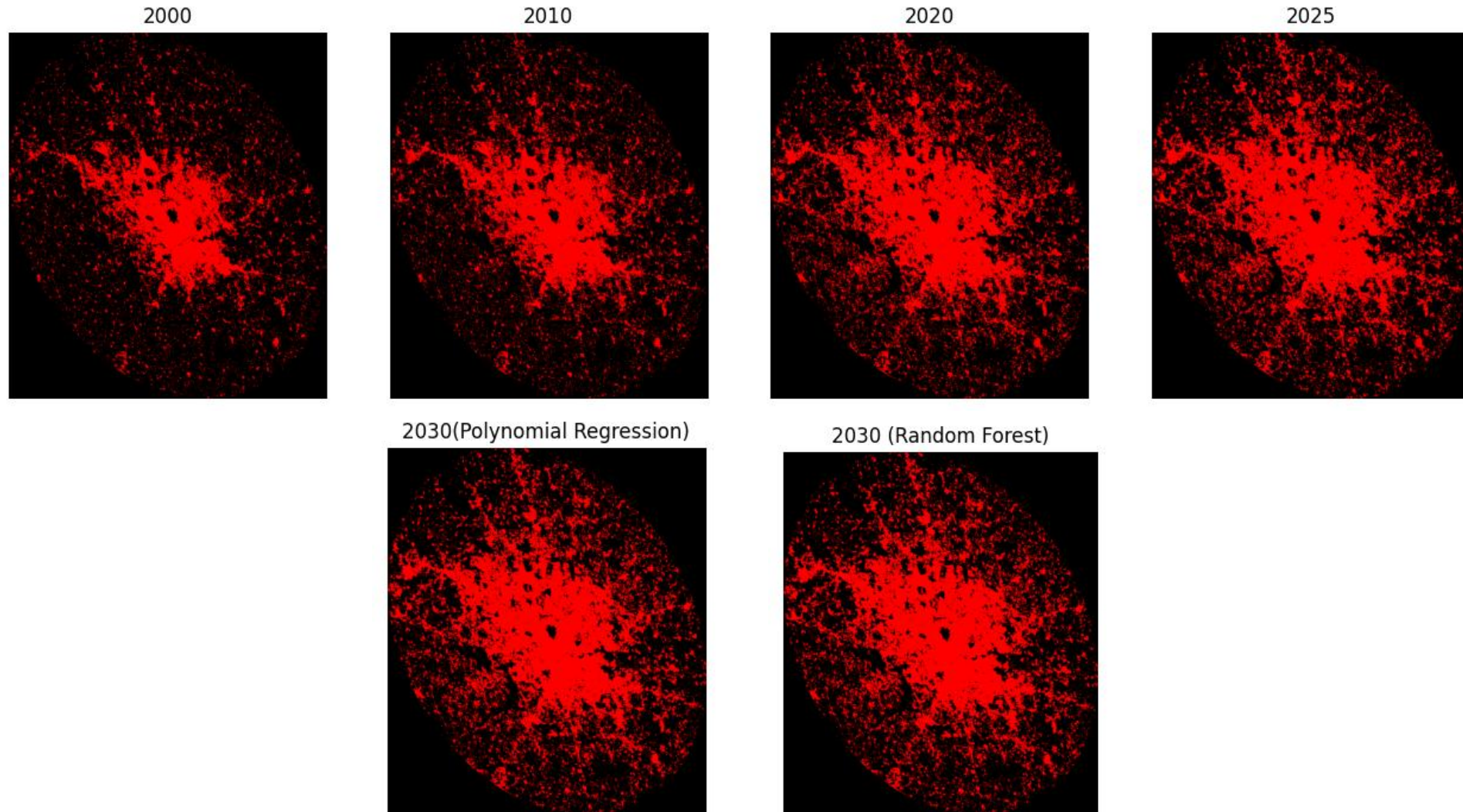


2030 (Random Forest)



11. RESULTS

Urban Sprawl in Hyderabad



13. REFERENCES

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Thank You